Enterprise Cloud Data Platform: Technical Architecture & Operational Runbooks

Confidential Technical Documentation - Production Tier-1 Environment

Section 1: Microservices Architecture & System Overview

The Enterprise Cloud Data Platform operates on an air-gapped, zero-cloud dependency architecture. The system decouples data ingestion, semantic vector indexing, hybrid retrieval, and generative inference into independent microservices. The ingestion subsystem utilizes IBM Docling for native layout-aware parsing of technical documents, preserving headers, tabular metadata, and code structures. Documents are segmented using recursive header chunking with a granular window of 550 characters and a 100-character overlap. The storage layer utilizes an embedded Qdrant vector database operating in local disk persistence mode. To achieve enterprise-grade recall, Qdrant combines dense vector embeddings generated by BAAI/bge-small-en-v1.5 (384 dimensions) with sparse BM25 token frequencies, fused via Reciprocal Rank Fusion (RRF).

Section 2: High Availability, Tier-1 SLA Metrics & Disaster Recovery

The platform guarantees high availability and strict service level agreements (SLAs) across all tier-1 production regions. The Tier-1 SLA specifications are as follows: Platform Uptime SLA is guaranteed at 99.995% monthly rolling uptime with tier-1 service credit penalties. Recovery Time Objective (RTO) is strictly under 15 minutes per incident event via automated standby cluster failover. Recovery Point Objective (RPO) is strictly 0 seconds (RPO=0) using synchronous multi-zone replication without data loss. P99 query latency SLA is under 120 ms under loads up to 15,000 requests per second. Cross-Encoder reranking latency SLA is under 85 ms across top-15 candidate evaluation.

Section 3: Core Incident Runbook: ERR-409-TIMEOUT

Error Code ERR-409-TIMEOUT: Distributed lock acquisition timed out after 30,000 ms. This error is triggered when an active transaction in the distributed worker cluster fails to acquire or renew a mutex in the distributed lock coordinator due to partition lag or high contention on shared tenant state. Incident Resolution Runbook: Step 1: Terminate the stale transaction thread immediately to prevent cascading deadlocks. Step 2: Inspect Redis cluster shard health and active lock keys using CLUSTER NODES and redis-cli --bigkeys. Step 3: Trigger an idempotent rollback of uncommitted transactional operations in Postgres. Step 4: Client applications MUST implement exponential backoff with randomized jitter between 200 ms and 5,000 ms before retrying.

Section 4: Security Compliance, Encryption & Zero-Trust Governance

The platform strictly adheres to SOC2 Type II, ISO/IEC 27001, and HIPAA compliance mandates for sensitive enterprise workloads. Data at rest stored within the embedded Qdrant vector database and attached NVMe block volumes is encrypted using AES-256-GCM. Cryptographic keys are managed via AWS KMS Customer Managed Keys (CMK) or HashiCorp Vault Enterprise with automated 90-day rotation. All internal network communications enforce mutual TLS (mTLS) with TLS 1.3 protocol standards, eliminating unencrypted plain-text transit. Role-Based Access Control (RBAC) and attribute-based security policies restrict data access at the individual document chunk level. Auditing logs are immutably streamed to an append-only WORM compliant storage bucket.

Section 5: Two-Stage Hybrid Retrieval & Cross-Encoder Calibration

To eliminate LLM hallucinations and optimize context precision, the RAG engine implements a two-stage retrieval pipeline. Stage 1 performs hybrid retrieval to fetch top-15 candidate chunks using Reciprocal Rank Fusion of BAAI/bge-small-en-v1.5 and BM25. Stage 2 routes these candidates through a local BAAI/bge-reranker-base cross-encoder model. The Cross-Encoder assigns probability scores directly in the range 0.0 to 1.0. An Abstention Guardrail is configured with a strict confidence threshold of 0.55. If the top-ranked chunk exhibits a rerank score below 0.55, the pipeline immediately abstains from generation, returning is_grounded=False and the message 'Insufficient context in the knowledge base to answer this question reliably.'

Section 6: Disaster Recovery Verification & Backup Retention Schedule

Disaster recovery (DR) simulations are executed bi-weekly against active-passive database replicas. Continuous WAL (Write-Ahead Logging) archiving ensures continuous point-in-time recovery to any millisecond within a 35-day retention window. Automated health checks ping the cluster control plane every 500 ms; if three consecutive probes fail, DNS traffic fails over automatically.